EAST WEST UNIVERSITY

Department of Computer Science and Engineering

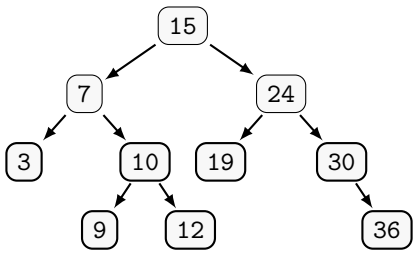
B.Sc. in Computer Science and Engineering Program

Class Test 2, Fall 2025

Course: CSE 207 Data Structures, Section-06 **Date:** November 30, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor
Full Marks: 10
Time: 20 Minutes

Note: There are **TWO** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

Consider the following Binary Search Tree (BST):



1. For the given BST above, **write down** the following traversal sequences:

(a) Pre-order traversal

(b) In-order traversal

(c) Post-order traversal

Clearly indicate the order of node visits for each traversal.

[CO2, C3, Mark: 5]

2. Starting from the **same BST** given above, **insert the key 11** following the standard Binary Search Tree insertion algorithm.

(a) Show the **sequence of node comparisons** during the insertion of 11.

(b) Draw the **resulting BST** after the insertion is complete.

[CO2, C3, Mark: 5]